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RGAnomaly: Data reconstruction-based generative adversarial networks for multivariate time series anomaly detection in the Internet of Things

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ABSTRACT

The Internet of Things encompasses a variety of components, including sensors and controllers, which generate vast amounts of multivariate time series data. Anomaly detection within this data can reveal patterns of behavior that deviate from normal operating states, providing timely alerts to mitigate potential serious issues or losses. The prevailing methodologies for multivariate time series anomaly detection are based on data reconstruction. However, these methodologies face challenges related to insufficient feature extraction and fusion, as well as instability in the reconstruction effectiveness of a single model. In this article, we propose RGAnomaly, a novel data reconstruction-based generative adversarial network model. This model leverages transformers and cross-attention mechanisms to extract and fuse the temporal and metric features of multivariate time series. RGAnomaly constructs a joint generator comprising an autoencoder and a variational autoencoder, which forms the adversarial structure with a discriminator. The anomaly score is derived from the combined data reconstruction loss and discrimination loss, providing a more comprehensive evaluation for anomaly detection. Comparative experiments and ablation studies on four public multivariate time series datasets demonstrate that RGAnomaly delivers superior performance in anomaly detection, effectively identifying anomalies in time series data within IoT environments.

1. Introduction

The continuous advancement of Internet of Things(IoT) technology has facilitated the interconnection of data between things, improving production and the efficiency of life [1]. It plays a pivotal role in areas such as smart cities, smart transportation, and smart industry. The IoT encompasses devices such as sensors, controllers, and control computers, which generate, collect, or monitor reflecting changes in the real world and the state of the devices [2,3]. This data is of significant strategic value for real-time monitoring and precise control within IoT systems. Nevertheless, in natural IoT environments, factors such as device failures, operational errors, omissions, or network attacks can cause IoT data to deviate from expected norms. These deviations may manifest as anomalous peaks and troughs or prolonged oscillations, deviating from normal operating patterns [4,5]. Data that differ significantly from regular observations are typically referred to as anomalies [6,7]. Anomaly detection is a technique for identifying such deviations, enabling timely alerts and responses to prevent anomalies from causing irreparable economic losses or personal injuries. For example, identifying anomalies in indicators such as PH levels, chlorophyll, dissolved oxygen, and heavy metals from water quality sensors enables the timely detection of changes in water pollution, thereby preventing damage to drinking water sources. Similarly, detecting anomalies in sensors and controllers within water treatment plants can promptly identify equipment failures, thereby avoiding safety issues and economic losses caused by anomalies in the water treatment process [8]. Consequently, anomaly detection in IoT data is of significant practical importance [9,10].

IoT data is collected by sensors and is characterized by distinct temporal and multivariate (or metric) features [11]. Consequently, this type of data is called multivariate time series data. Due to the unknown nature of anomalies, significant variations across domains, and the lack of labeled data, anomaly detection in multivariate time series data often employs unsupervised learning methods, which are more adaptable and practical than supervised learning methods. Concerning model complexity, traditional machine learning methods are inadequate for handling complex and dynamically changing data anomalies. While certain traditional machine learning models, such as Hidden Markov Models and Dynamic Bayesian Networks, can capture temporal or

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metric correlations to some extent, they face limitations in dealing with high-dimensional and complex non-linear data patterns. Consequently, unsupervised deep learning methods have emerged as the mainstream approach for anomaly detection in multivariate time series data. Deep learning approaches can be mainly categorized into data reconstruction-based and data prediction-based methods. Reconstruction-based methods learn the distributional characteristics of multivariate time series data to reconstruct the data, identifying anomalies by comparing the differences between the reconstructed data and the original data [12–14]. In this approach, the model seeks to minimize the discrepancies between the original data and its reconstruction. In contrast, due to their distinctive distributional characteristics, anomalies exhibit more pronounced discrepancies between the reconstructed and original data. Prediction-based methods detect anomalies by forecasting time series values and comparing the predicted series with the actual series [15–17]. However, certain data features, such as metric features, are challenging to predict in complex IoT environments. Furthermore, recent studies have also explored virtual redundancy models that leverage relationships among time series for anomaly detection [18], as well as digital twin models that analyze error probability distributions between model outputs and actual values to detect anomalies [19]. Nevertheless, considering various approaches comprehensively, reconstruction-based anomaly detection methods have emerged as a significant research direction for multivariate time series anomaly detection due to their straightforward and effective approach as well as their outstanding detection performance.

Although current data reconstruction-based anomaly detection models have achieved some success, there are still issues in feature extraction, fusion, and data reconstruction. Multivariate time series data exhibit distinct temporal features and originate from different sensors, thus exhibiting metric features. Existing methods typically consider temporal or metric features, neglecting their implicit correlations. In the data reconstruction process, existing methods employ basic network structures such as Autoencoder(AE), Variational Autoencoder(VAE), or Generative Adversarial Network(GAN) and generally utilize only a single type of network structure for data reconstruction [20–22]. Each of these single network structures has inherent limitations. For instance, the AE structure learns local features of time series data, which may not adequately reconstruct unseen data. Although the VAE structure introduces variational latent variables, providing some capability to handle unknown anomalies, its reconstruction results can be more blurred than AE, ignoring local variable features [23]. Conversely, GAN suffers from training difficulties, gradient vanishing, and mode collapse issues [24]. These limitations result in single reconstruction models exhibiting good reconstruction capabilities for some multivariate time series data while struggling with others, leading to unstable reconstruction results.

To address the issues mentioned above, this article proposes a novel data reconstruction-based generative adversarial network for anomaly detection in multivariate time series data within the IoT. The network includes a unique feature extraction and fusion module, which is designed as a dual-transformer architecture combined with a cross-attention mechanism to extract and integrate temporal and metric features. The transformer can capture global dependencies and correlations from both temporal and metric features. Meanwhile, the cross-attention mechanism establishes correspondences between temporal and metric features, enabling the alignment, association, and efficient fusion of information from the two feature spaces. This module effectively addresses the limitations of existing models in feature extraction and feature fusion. A generative adversarial network module then processes the extracted and fused features. This module draws inspiration from the ensemble concept of weak classifiers in machine learning, addressing the instability of single data reconstruction models by combining multiple reconstruction models for adversarial generation. This module integrates AE and VAE as a generator. The unique generator combines the reconstruction capabilities of AE with the distribution modeling capabilities of VAE, enabling the model to detect not only explicit anomalies but also subtle deviations or unknown anomalies by capturing complex patterns and uncertainties in the data. Furthermore, the generative adversarial structure of the module significantly enhances the data reconstruction capability of the generator. By jointly optimizing the reconstruction loss of the generator and the discrimination loss of the discriminator, the model can effectively and comprehensively detect anomalies.

In summary, the contributions of this paper are as follows:

1. We propose RGAnomaly, a data reconstruction-based generative adversarial network that effectively detects anomalies in multivariate time series data from the IoT.
2. The RGAnomaly model incorporates a distinctive feature extraction module and GAN structure. This architectural design effectively extracts and fuses the temporal and metric features of time series data, employing the GAN module to integrate multiple data reconstruction models for anomaly detection. This design enhances data reconstruction's stability and anomaly detection's effect.
3. We conducted comparative experiments on four public multivariate time series datasets and performed ablation studies on the different modules of the RGAnomaly model. The experimental results demonstrate the superiority and effectiveness of the proposed RGAnomaly model. In addition, we performed visual analysis and sensitivity analysis, further refining the model's performance evaluation.

2. Related work

The multivariate time series anomaly detection method based on data reconstruction detects anomalies by analyzing the residuals between the reconstructed sequence and the original sequence. The primary challenge lies in accurately reconstructing the original data. Because Long Short-Term Memory (LSTM) networks [25] can capture temporal trends in time series, the LSTM-AE model uses an autoencoder to model normal data, determining anomalies by comparing reconstruction errors [26]. Similarly, Gated Recurrent Unit (GRU) as a simplified version of LSTM, GRU-AE can be used for data reconstruction with reduced computational complexity [27]. The LSTM-VAE model employs a variational autoencoder based on LSTM to detect anomalous actions in robot signals, also VAE provides data generation capability compared to AE. OmniAnomaly extends LSTM-VAE by incorporating techniques such as random variable connection and planar normalization to capture normal patterns in multivariate time series data [28]. MAD-GAN utilizes an LSTM-RNN GAN structure to detect anomalies in IoT data generated by Cyber-Physical Systems (CPS) [29]. TanoGAN features a network structure very similar to MAD-GAN [30]. Since the GAN is highly dependent between the generator and the discriminator in the training phase, in order to reduce this dependency, the GAN can be trained by adding additional training steps [31]. AMBi-GAN constructs a GAN based on bidirectional LSTM and attention mechanisms, effectively capturing temporal dependencies in time series [32]. Since these data reconstruction models have certain drawbacks, some researchers have begun to integrate multiple models. RANSynCoder employs a set of AEs to detect anomalies, where each AE's input is a randomly selected subset of features that contain sufficient information to reconstruct the entire sequence, and anomalies are determined through a voting mechanism [33]. USAD combines AE and GAN for time series anomaly detection by sharing an encoder between two AE models to construct a GAN, with the autoencoder structure improving GAN training stability [34]. The DAEMON model combines AE and GAN to detect anomalies in time series [35]. TMANomaly enhances data reconstruction capability by constructing two identical TRecAE sub-networks that switch roles between generator and discriminator, engaging in adversarial training [13]. It is worth mentioning that TMANomaly also employs Grey Relation Analysis (GRA) to select dataset features. Similarly, the use of dimensionality reduction methods for multidimensional

time series has the potential to reduce complexity, denoise, and retain essential information for anomaly detection [36]. From the perspective of data reconstruction, most current anomaly detection methods rely on a single data reconstruction model, which struggles to handle unknown and dynamically complex anomalies effectively. While models such as RANSynCoder and USAD combine AE and GAN for data reconstruction, the AE's lack of generative capacity limits the model's ability to learn the data distribution, and using similar types of reconstruction models has inherent limitations.

Most current anomaly detection models focus on temporal features in the feature extraction, using network structures like LSTM. However, in IoT environments, sensor devices generate vast amounts of multivariate time series data containing very complex temporal patterns. The Transformer [37], with its self-attention mechanism, can model complex temporal patterns in time series. The GTA model introduces a multi-branch attention mechanism Transformer structure, combining global attention, multi-head self-attention, and neighborhood convolution to replace the conventional multi-head self-attention mechanism [38]. TranAD is a Transformer-based anomaly detection model with a structure similar to USAD. It involves a two-stage training process with AE and GAN structures, using self-regulation and adversarial training to amplify errors and achieve training stability [39]. ADTransformer adopts a Transformer adversarial network structure similar to TranAD and USAD, proposing new anomaly score smoothing and prior knowledge strategies [40]. STL-transformer performs seasonal, trend, and residual decomposition of the time series and uses a one-dimensional convolutional network to capture the local dependencies of the time series [41]. Incorporating a trend estimation-based test time strategy into the Transformer can enhance the model's temporal perception for time series [42]. Despite the improved temporal feature extraction using the Transformer structure, the extraction of metric features and their integration with temporal features have been neglected. InterFusion [43], an unsupervised anomaly detection method, models normal patterns in multivariate time series data using a hierarchical VAE with two stochastic latent variables, each learning low-dimensional temporal embeddings, and inter-metric embeddings, thus considering both temporal and metric characteristics in time series. However, InterFusion employs a two-view embedding strategy to extract temporal features and then uses an SRNN-like structure to further learn metric features from the extracted temporal features. Although this ensures temporal consistency within metric features, it can lead to feature loss and feature imbalance issues. Therefore, capturing and associating temporal and metric features remain critical unresolved challenges for multivariate time series data.

3. Problem definition

The core of data reconstruction-based anomaly detection methods lies in reconstructing the input data and evaluating based on the differences between the reconstructed data and the original data [44]. In the real world, there is a significant imbalance between normal and anomalous data, with normal data far outnumbering anomalous data. Furthermore, multivariate time series anomaly detection involves detecting anomalies in unlabeled time series data, which requires learning informative representations and establishing distinguishable criteria [45]. Therefore, data reconstruction-based anomaly detection methods typically learn from normal data to understand its characteristics or distribution. During detection, the reconstruction residuals for normal data are relatively small, while anomalous data, which does not conform to the characteristics or distribution of normal data, results in larger reconstruction differences.

Consider a multivariate time series dataset $X \in \mathbb{R}^{T\times N}$ sourced from the IoT.

$$X = \left\{x_1^N, x_2^N, \ldots, x_t^N, \ldots, x_T^N\right\} \tag{1}$$

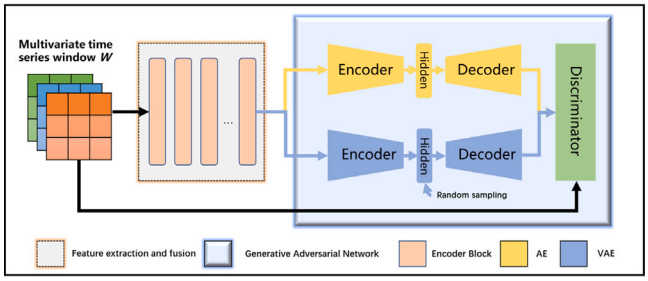


Fig. 1. The complete structure of the RGAnomaly. The model includes a feature extraction fusion module and a generative adversarial network. The feature extraction fusion module is composed of unique encoder blocks. The generative adversarial network includes AE, VAE, and discriminator.

Here, T represents the length of the time series, N represents the number of metric features observed at each time point, and x_t^N denotes the observation at any given time t. For convenience, we use X to denote the training data. Similarly, any unseen test time series is denoted as $\hat{X} \in \mathbb{R}^{\hat{T}\times N}$, where $\hat{T}$ is length of the test series, and its dimension remains consistent with X. During testing, the model reconstructs the test data (with the reconstructed data denoted as $\hat{X}'$) and calculates the anomaly score at each time point by comparing $\hat{X}'$ with the original test data $\hat{X}$. The anomaly score is then compared against a threshold δ to determine the anomaly detection results $Y = \left\{y_1, \ldots, y_t, \ldots, y_{\hat{T}}\right\}$, where each y_t is a binary variable: $y_t \in (1, 0)$, indicating the anomaly state at the corresponding time point.

4. Method

The overall structure of RGAnomaly is illustrated in Fig. 1. The model employs transformers to extract temporal and metric features from multivariate time series data, which are subsequently fused through a cross-attention mechanism to generate a unified feature matrix. The generator of the GAN consists of a joint model of an AE and a VAE, which performs data reconstruction on the feature matrix. The reconstructed data, along with the original data, are fed into the discriminator of the GAN, which discerns whether the data are actual or reconstructed. This integrated GAN structure enhances the data reconstruction capability of the anomaly detection model, thereby improving its anomaly detection performance.

4.1. Data preprocessing

To facilitate better data detection, we performed data preprocessing on the time series data. First, we cleaned the datasets by removing irrelevant unique attributes (e.g., device ID) and applied forward filling to handle missing values. Next, we normalized the dataset using min–max normalization as:

$$x_t^N = \frac{x_t^N - \min(X)}{\max(X) - \min(X)} \tag{2}$$

This normalization method performs a linear transformation on the time series data, mapping the values to the range [0, 1].

Finally, we applied a sliding window operation to the dataset. The sliding window is a widely used preprocessing method for time series data that generates segments containing consecutive time steps. Compared to individual time step vectors, the sliding window approach enables the model to learn the contextual dependencies between time steps, capture more complex temporal patterns. Consequently, we apply a sliding window to partition X and $\hat{X}$ with a window size of w. Taking time series X as an example, the sliding window sequence from time $t - w + 1$ to time t is as follow.

$$W_{t-w+1:t} = \left\{x_{t-w+1}^N, \ldots, x_{t-1}^N, x_t^N\right\} \tag{3}$$

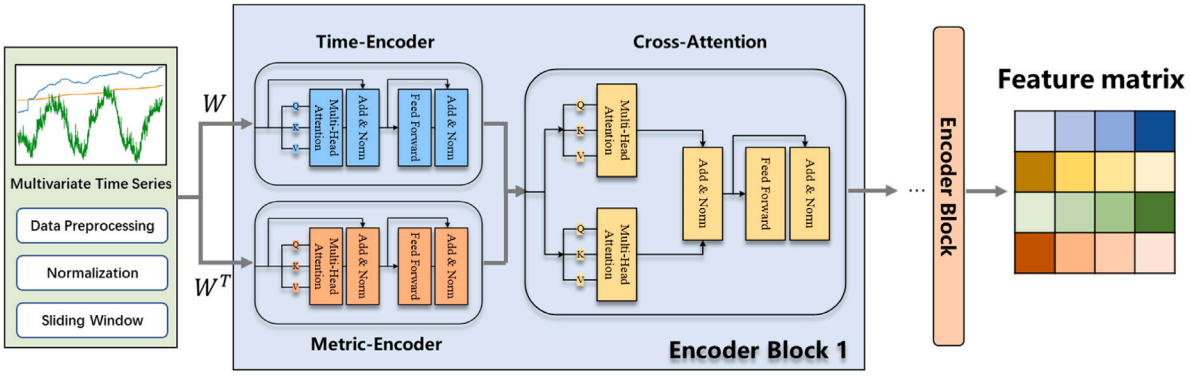


Fig. 2. The feature extraction fusion process of the RGAnomaly. The model extracts the time feature and the metric feature of the data and carries on the cross-fusion to obtain the feature matrix C.

By continuously sliding the window, we obtain the complete window sequences $W = \{W_{1:w}, \ldots, W_{t-w+1:t}, \ldots, W_{T-w+1:T}\}$, which represents the training data window sequences. Similarly, the test data window sequences are obtained in the same manner on $\hat{X}$ and are denoted as $\hat{W}$. Thus, during both training and testing, the input data take the form of W or $\hat{W}$.

4.2. Feature extraction and fusion

The unique Encoder Block is the core of RGAnomaly's feature extraction and fusion, as shown in Fig. 2. The multivariate time series data undergoes preprocessing, normalization, and sliding window segmentation to obtain the time window sequence W and its transpose, which are then input into the Encoder Block.

To fully extract the temporal and metric features in the time series data, two structurally identical transformer encoders form a dual transformer structure: time-encoder and metric-encoder. These encoders leverage self-attention mechanisms to learn attention features in the temporal and metric dimensions, respectively. Consistent with the classic Transformer, we define the corresponding matrices W_z^q, W_z^k and W_z^v for different encoders, where z represents either time(t') or metric(m'). The time window sequence undergoes matrix operations to obtain the Query (Q_z), Key (K_z), and Value matrices (V_z), and then a multi-head attention mechanism is applied. Here, the $Attention()$ function represents the computation method for any single head within a multi-head attention mechanism, and $\sqrt{d}$ is a scaling factor that can reduce the variance of weights, thus preventing the inner product from becoming too large.

$$Q_z, K_z, V_z = WW_z^q, WW_z^k, WW_z^v \tag{4}$$

$$\text{Attention}_z\left(Q_z, K_z, V_z\right) = softmax\left(\frac{Q_z {K_z}^T}{\sqrt{d}}\right)V_z \tag{5}$$

After computing the attention matrices for all heads, they are concatenated to obtain the complete multi-head attention matrix. Then passing through two *Add&LayerNorm* layers, the multi-head attention matrix yields the feature encoding matrix C_z corresponding to the temporal and metric features.

We achieve deep fusion of the two feature encoding matrices through a cross-attention mechanism. In cross-attention, after matrix transformations similar to those described previously, C_z yield their respective Q_z^c, K_z^c and V_z^c matrices. The superscript c denotes the cross-attention stage, distinguishing it from the previous attention stage. During the attention computation, the query vector $Q_{t'}^c$ from the temporal dimension is used as the query input to compute with $K_{m'}^c$ and $V_{m'}^c$ from the metric dimension. This method introduces the key–value information of the metric dimension into the query information of the temporal dimension, thus achieving cross-attention. Similarly, the metric dimension's query information will incorporate the temporal dimension's key–value information. For example, in the temporal dimension, the cross-attention calculation is as follows:

$$Attention_{t'}^c(Q_{t'}^c, K_{m'}^c, V_{m'}^c) = softmax(\frac{Q_{t'}^c {K_{m'}^c}^T}{\sqrt{d}})V_{m'}^c \tag{6}$$

The cross-attention matrices are concatenated and then passed through similar *Add&LayerNorm* layers for residual computation, addressing the training issues of deep networks. This results in the feature matrix C with deeply fused encodings. Matrix C contains temporal and sensor metric features and the interactions between them. Since the encoder block can be a single unit or a combination of multiple units, the feature matrix C can be used directly for data reconstruction or as input to the next encoder block, iterating the above process. While multiple Encoder Blocks can provide deeper feature extraction and fusion, they can also disrupt feature continuity. Therefore, in our experiments, we tested different numbers of layers for the blocks. Here, we refer to the final feature encoding matrix uniformly as C.

4.3. Data reconstruction-based generative adversarial networks

In order to overcome the limitations of a single model in data reconstruction, RGAnomaly consists of two independent models, AE and VAE, to reconstruct the feature encoding matrix C. AE pays more attention to the local characteristics of time series data. In comparison, VAE pays more attention to the global data distribution of time series data than AE due to the addition of variational hidden variables. Two distinct data reconstruction models can reconstruct data from different perspectives, avoid the limitations of a single data reconstruction model, and improve the effect of data reconstruction.

In RGAnomaly, the encoder and decoder of both AE and VAE consist of fully connected layers. The AE performs a linear layer transformation on the matrix C to obtain the hidden layer space Z_{AE}, and then the decoder reconstructs the hidden layer space to obtain the data W'_{AE}. The VAE similarly performs a linear layer transformation on the matrix C to obtain the hidden layer space Z_{VAE} and then performs random sampling to reconstruct the data W'_{VAE}.

RGAnomaly forms a generative adversarial network structure by adding a discriminator to improve the data reconstruction ability of AE and VAE. Under this structure, the data reconstruction part composed of AE and VAE, as the generator of GAN, generates two different kinds of reconstructed data, which are input into the discriminator together with the original data. The discriminator determines whether the input data is actual or reconstructed from the data.

When calculating the loss function, the loss function of the feature encoding module during model training is derived from the weighted sum of the reconstruction losses of AE and VAE, as both AE and VAE perform reconstruction on the matrix C. The reconstruction loss of AE

is the $L2$ norm between the reconstructed data and the original data, while the reconstruction loss of VAE includes not only the reconstruction error but also the Kullback–Leibler (KL) divergence of the data distribution, where p represents the standard normal distribution and q is the approximate distribution of the model:

$$Loss(AE) = \|W'_{AE} - W\|_2 \tag{7}$$

$$Loss(VAE) = \|W'_{VAE} - W\|_2 \ + \ D_{KL}(q \parallel p) \tag{8}$$

$$Loss(Encoder) = Loss(AE) + Loss(VAE) \tag{9}$$

With regard to the generative adversarial network component of the model, the loss function of the GAN is as follows:

$$\begin{aligned} Loss(GAN) = &\log(Dis(W)) + \log(1 - Dis(W'_{AE})) \\ &+ \log(1 - Dis(W'_{VAE})) \end{aligned} \tag{10}$$

The symbol $Dis()$ denotes the probability that the discriminator discriminates as true. During training, the model expects the loss function of the discriminator to be gradient ascending, which proves that the discriminator can distinguish the actual data from the reconstructed data well. In contrast, the data reconstruction module composed of AE and VAE, as the generator of the generative adversarial network, needs to minimize the loss function of GAN to prove that the reconstructed data can effectively confuse the discriminator. Therefore, the loss functions of AE and VAE also include the corresponding discriminator loss, which enhances the data reconstruction capability through gradient descent. Through iterative updates, AE and VAE can learn the characteristics and distribution of normal data. Consequently, they produce minor reconstruction errors when dealing with normal data and more significant errors when reconstructing anomalous data. Meanwhile, the discriminator in the GAN, having learned the distribution patterns of the original normal data, can accurately evaluate the anomalies during detection. Thus, RGAnomaly achieves more accurate discrimination of anomalous data, avoiding the limitations of a single model that may only be able to handle some types of anomalies effectively. The training process is summarized in Algorithm 1.

Algorithm 1 The RGAnomaly training algorithm

Input: Normal windows Dataset W, Number epochs N.
Output: Trained Encoder, AE, VAE and GAN's discriminator.

Encoder,AE,VAE and discriminator ← initialize weights
$n \leftarrow 1$
while $n \leq N$ **do**
  for t=1 to Length(W) **do**
    Feature matrix C ← Encoder(W_t)
    $W'_{AE} \leftarrow$ AE(C)
    $W'_{VAE} \leftarrow$ VAE(C)
    $Loss(AE) \leftarrow ||W'_{AE} - W_t||_2$
    $Loss(VAE) \leftarrow ||W'_{AE} - W_t||_2 + D_{KL}$
    $Loss(Encoder) \leftarrow Loss(AE) + Loss(VAE)$
    Input (W_t,W'_{AE},W'_{VAE}) → GAN's discriminator.
    Calculate the probability that the discriminator is true using $Dis()$.
    $Loss(GAN) \leftarrow \log(Dis(W)) + \log(1 - Dis(W'_{AE})) + \log(1 - Dis(W'_{VAE}))$
    $Loss(AE)$ and $Loss(VAE)$ plus corresponding $Dis()$ losses.
    Encoder,AE,VAE and discriminator ← update weights using lossess.
  end for
  $n \leftarrow n$+1
end while

4.4. Anomaly evaluation criteria

We propose an assessment criterion that integrates data reconstruction and discriminative scores during anomaly evaluation. The reconstruction score is a weighted sum of the reconstruction scores from AE and VAE. In real-world scenarios, it is unclear whether anomalies are more likely to deviate from data characteristics or data distribution. Therefore, the reconstruction scores from AE and VAE are given equal weights, denoted as $score(Rec)$. Let $\hat{W}$ represent the test window.

$$score(Rec) = \|AE(\hat{W}) - \hat{W}\|_2 + \|VAE(\hat{W}) - \hat{W}\|_2 \tag{11}$$

The discrimination score is calculated using the discriminator to identify the input data. The discrimination score $score(Dis)$ is as follows:

$$score(Dis) = 1 - Dis(\hat{W}) \tag{12}$$

In practical applications, the input data is susceptible to anomalous moments. This paper adopts a distinguishing strategy of fusing abnormal probability [40], which is calculated based on the sigmoid mapping score of each time moment within the time series window. The probability and mapping score are then combined to yield an anomaly score, denoted as $Score(L)$, where L stands for Rec or Dis.

$$Score(L) = Softmax(R(L)) \odot R(L) \tag{13}$$

$$R(L) = Sigmoid(score(L)) \tag{14}$$

In this way, the influence of abnormal data on normal time can be reduced, and the score of abnormal time can be expanded. The above operations are performed on both the reconstruction score and the discrimination score, and the final Anomaly Score is the weighted sum of them(α is the weight parameter):

$$AnomalyScore = \alpha Score(Rec) + (1 - \alpha) Score(Dis) \tag{15}$$

Assuming that the anomaly threshold is δ, the following expression is used to determine whether $\hat{W}$ is abnormal. When the result is 1, it is abnormal:

$$result = \begin{cases} 1, & Anomaly\ Score > \ \delta \\ 0, & Anomaly\ Score \leq \ \delta \end{cases} \tag{16}$$

5. Experimental setup

5.1. Datasets

To assess the efficacy of the RGAnomaly model, we selected four public datasets widely used for multivariate time series anomaly detection. These datasets originate from the IoT or related CPS, sensor devices, and control nodes.

Secure Water Treatment (SWaT) Dataset [8]: The SWaT dataset is a collection of data derived from a natural industrial water treatment plant's testbed, which simulates the water treatment process. The dataset includes real-time data collected from 51 units comprising sensors and controllers.

Water Distribution (WADI) Dataset [46]: The WADI dataset represents an extension of the SWaT testbed. The dataset encompasses the complete process of water treatment, distribution, and storage, containing 127 feature dimensions (123 features are from sensors and controllers in the water distribution process).

Mars Science Laboratory (MSL) Dataset [26]: The MSL dataset, annotated by NASA experts, includes sensor telemetry data from the Curiosity rover for anomaly detection. The dataset comprises 55 feature dimensions.

Pooled Server Metrics (PSM) Dataset [33]: The PSM is a control node server metric dataset published by eBay, which contains 25 server collection metrics.

Table 1
Comparison of experimental results between the RGAnomaly and baselines on four public datasets.

Model	SWaT			WADI			MSL			PSM		
Metric	P	R	F1	P	R	F1	P	R	F1	P	R	F1
LSTM-AE	0.8615	0.8327	0.8469	0.8143	0.8482	0.8309	0.8545	0.8250	0.8395	0.7693	0.8964	0.8280
LSTM-VAE	0.7600	0.8950	0.8220	0.9897	0.6377	0.7756	0.8549	0.7994	0.8262	0.7362	0.8992	0.8096
MAD-GAN	0.9009	0.7834	0.8380	0.2233	0.9124	0.3588	0.5403	0.8779	0.6689	0.6800	0.9007	0.7750
BeatGAN	0.6401	0.8746	0.7392	0.7048	0.7226	0.7136	0.8975	0.8542	0.8753	0.9030	0.9384	0.9204
USAD	0.9851	0.6618	0.7917	0.6451	0.3220	0.4296	0.7949	0.9912	0.8822	0.6770	0.9012	0.7732
TranAD	0.8901	0.7834	0.8333	0.3529	0.8296	0.4951	0.5398	0.8302	0.6542	0.9005	0.8570	0.8782
OmniAnomaly	0.8142	0.8430	0.8283	0.9825	0.6497	0.7822	0.8902	0.8637	0.8767	0.8839	0.7446	0.8083
Interfusion	0.8059	0.8558	0.8301	0.8026	0.9100	0.8529	0.8128	0.9270	0.8662	0.8361	0.8345	0.8353
CL-MPPCA	0.7678	0.8150	0.7907	0.7096	0.7521	0.7302	0.7371	0.8854	0.8044	0.5602	0.9993	0.7180
MGUAD	0.9851	0.7204	0.8332	0.9281	0.4114	0.5701	0.9100	0.8689	0.8890	0.9990	0.7913	0.8831
TMANomaly	0.9417	0.8042	0.8675	0.9501	0.6998	0.8060	0.8815	0.9032	0.8922	0.9822	0.8638	0.9192
AT-DCAEP	0.9977	0.6770	0.8066	0.7880	0.8295	0.8082	0.9133	0.8880	0.9004	0.9323	0.9340	0.9331
ADTransformer	0.9890	0.7618	0.8607	0.8007	0.6068	0.6904	0.9200	0.8864	0.9029	0.9934	0.8647	0.9246
TTA-transformer	0.8949	0.8672	0.8808	0.9204	0.7239	0.8104	0.9169	0.8980	0.9074	0.9834	0.9335	0.9578
STL-transformer	0.9732	0.8277	0.8946	0.8468	0.8626	0.8546	0.9137	0.8441	0.8775	0.9471	0.9826	0.9645
RGAnomaly(ours)	0.9452	0.8540	**0.8973**	0.9196	0.8139	**0.8635**	0.9268	0.9224	**0.9246**	0.9889	0.9665	**0.9775**

Boldface represents the optimal result and the underline represents the sub-optimal result.

The four datasets utilize a five-fold cross-validation approach during training, where the original training set is divided into five mutually exclusive subsets of similar size. For each training iteration, 80% of the data is selected as the training set, while the remaining 20% is used as the validation set. To ensure temporal consistency in the time series data, a sliding window operation is first applied to the original training set, segmenting the data into fixed-length time windows. Cross-validation is then performed on these segmented time windows, preserving the temporal consistency within each time series window.

5.2. Evaluation criteria and baselines

In terms of evaluation criteria, we use precision, recall, F1 score, false positive rate (FPR) and false negative rate (FNR) to evaluate the performance of anomaly detection models:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}\ ,\ \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}\ ,\ \mathrm{F1} = 2\cdot\frac{\mathrm{P}\cdot\mathrm{R}}{\mathrm{P}+\mathrm{R}} \tag{17}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TN}+\mathrm{FP}}\ ,\ \mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{TP}+\mathrm{FN}} \tag{18}$$

In this context, P and R represent precision and recall, respectively. TP represents true positive samples, FP represents false positive samples, TN represents true negative samples and FN represents false negative samples. In the evaluation, we adopt the point adjustment strategy commonly used in anomaly detection [28,47]. That is when detecting a segment of continuous anomalies, as long as part of the continuous anomalies is correctly detected, the entire abnormal segment is considered correctly detected.

The following models are selected as baselines: LSTM-AE [26], LSTM-VAE [12], MAD-GAN [29], BeatGAN [48], USAD [34], TranAD [39], OmniAnomaly [28], MGUAD [49], Interfusion [43], CL-MPPCA [50], TMANomaly [13], AT-DCAEP [51], ADTransformer [40], TTA-transformer [42] and STL-transformer [41]. These models have demonstrated effective anomaly detection in recent years. However, they present challenges in feature extraction and fusion and in integrating multiple reconstruction models.

5.3. Implementation

In terms of model implementation, we initially process the data using a non-overlapping sliding window approach across all datasets, with the window size set to 50. About our proposed RGAnomaly model, we configure the EncoderBlock with a single layer, set the number of attention heads to 1, and uniformly set the dimension of the Transformer embedding vectors d_model to 512. The data reconstruction components, AE and VAE, utilize a simple three-layer linear model for the encoder and decoder. The encoder consists of an input layer, a middle layer, and a hidden layer, while the decoder consists of a hidden layer, a middle layer, and an output layer. The dimensions of the middle layers are 512, and the hidden layers are 256. Similarly, the GAN's discriminator employs a three-layer linear model comprising an input, hidden, and output layer, with the hidden layer dimension set to 256. In the training process, the dataset's batch size is uniformly set to 256, and the learning rate is set to 0.001 by default. Adam [52] is utilized as the optimizer, and StepLR is used for dynamic learning rate adjustment. The anomaly threshold, denoted by δ, is determined based on the anomaly ratio r. Specifically, we compute anomaly scores for all time points on the validation set and select the score corresponding to the anomaly ratio r as the threshold, which enables the r proportion of data to be marked as anomalous [40]. Based on common settings, for the SWaT dataset, the anomaly ratio r is 0.1%. In contrast, r is 1% for the other datasets. The corresponding time point is marked as anomalous if the anomaly score exceeds the threshold δ. All experiments are implemented by PyTorch [53] with a single NVIDIA GeForce RTX 3090.

5.4. Experiments

5.4.1. Comparative experiments

Experiments were conducted with RGAnomaly and other baselines on four public datasets, and the corresponding P, R and F1 values were obtained. The experimental results are shown in Table 1.

The comparative experiments demonstrate that RGAnomaly achieves the best results across all four public datasets. As an excellent anomaly detection model for last year, ADTransformer extracts complex temporal features through the Transformer structure and obtains suboptimal anomaly detection effects on one dataset through the anomaly score smoothing strategy and prior knowledge strategy. Compared to ADTransformer, RGAnomaly demonstrated an improvement in F1 score of approximately 2.9% on average for the SWaT and MSL datasets, 5.3% on the PSM dataset, and 17.3% on the WADI dataset. The STL-Transformer, which decomposes time series data and integrates convolutional layers for data reconstruction, has proven to be an exceptionally effective anomaly detection model this year. Compared to the STL-Transformer, RGAnomaly achieved improvements of 0.3% on the SWAT dataset, 0.9% on the WADI dataset, 4.7% on the MSL dataset, and 1.3% on the PSM dataset. TTA-Transformer addresses the problem of time series anomaly detection by introducing a test-time adaptation strategy based on trend estimation and is one of the latest state-of-the-art (SOTA) methods. Compared to TTA-Transformer, RGAnomaly achieves an average improvement of approximately 2.7% across the four datasets. Similarly, AT-DCAEP is also one of the latest anomaly detection methods, and RGAnomaly

achieves an average improvement of 5.4% over it across the four datasets. TMANomaly, an exceptional anomaly detection model from last year, employs a reconstruction-based structure similar to USAD and TranAD and benefits from integrating grey relational analysis, resulting in relatively better detection performance. Nevertheless, RGAnomaly outperforms TMANomaly with an average improvement of 4.4% across the four datasets. In addition, we calculated the average FPR and average FNR of the aforementioned models across the four datasets, as presented in Table 2. From Table 2, it can be observed that our model achieved the sub-optimal result in terms of FPR, with only a slight gap compared to the best-performing model, while achieving the best result in terms of FNR. FPR represents the rate of normal instances being misclassified as anomalies, which has relatively minor implications in real-world scenarios since false alarms can typically be identified and corrected by human operators. In contrast, FNR reflects the rate of missed detections, where anomalies are misclassified as normal. The consequences of missed detections are more serious compared to false alarms, as undetected anomalies may lead to system malfunctions or sensor failures, resulting in potentially irreparable damage. Considering the comprehensive performance on both FPR and FNR, our model demonstrates the best error rate performance. Overall, RGAnomaly demonstrates superior performance compared to these latest anomaly detection methods.

Returning to Table 1, the optimal and sub-optimal results are proved that Transformer-based models, compared to recurrent neural network models like LSTM, better capture complex temporal features from time series data, resulting in higher average F1 scores. Additionally, we can observe that many models in the table, such as BeatGAN, OmniAnomaly and MGUAD, exhibit performance instability, performing well on some datasets but poorly on others. These experimental results highlight the limitations of single reconstruction models, as they struggle to meet the detection needs of datasets with varying data distributions. In contrast, RGAnomaly overcomes this issue by integrating multiple data reconstruction models and utilizing GAN to continuously improve the reconstruction capability of all models. This approach amplifies the differences between anomalous and normal data, thereby enhancing anomaly detection performance. The WADI dataset has the lowest average F1 score among the four datasets, likely due to its higher number of metric feature dimensions compared to the other datasets. RGAnomaly can learn temporal and metric features simultaneously, leading to better performance on the WADI dataset. Interfusion is also capable of extracting temporal and metric features, which contributes to its relatively good performance on the WADI dataset. However, Interfusion performs feature learning in Two-view embedding: it first focuses on temporal features and then extracts metric features from them. This sequential process assigns different priorities to temporal and metric features during the learning phase, potentially leading to feature loss or feature imbalance. In contrast, RGAnomaly simultaneously extracts temporal and metric features, reducing the risk of information loss. Moreover, it employs cross-attention mechanisms for feature fusion, enabling it to better capture the interactions between temporal and metric features. Experimental results demonstrate that our model significantly outperforms Interfusion. In conclusion, the comparative experimental results demonstrate the RGAnomaly model's superior performance in the anomaly detection task.

5.4.2. Ablation studies

To verify the effectiveness of each module of the proposed RGAnomaly, we conducted ablation studies. For data feature extraction and fusion, we obtain two ablation models, RGAnomaly-onlyTime and RGAnomaly-onlyMetric. The RGAnomaly-onlyTime model extracts only temporal features during feature extraction, while the RGAnomaly-onlyMetric model similarly extracts metric features. By dividing the data reconstruction module and the generative adversarial network, we obtain three ablation experimental models, namely RGAnomaly-onlyAE, RGAnomaly-onlyVAE, and RGAnomaly-noGAN, which remove VAE, AE, and GAN, respectively. The results of ablation studies are shown in Table 3.

Table 2
Average FPR and FNR of different models.

Model	Average FPR(%)	Average FNR(%)
TMANomaly	0.72	18.22
AT-DCAEP	1.16	16.79
ADTransformer	**0.48**	22.01
TTA-transformer	0.81	14.44
STL-transformer	1.02	12.08
RGAnomaly	0.57	**11.08**

Table 3
Ablation studies of different modules of the RGAnomaly.

Model	SWaT	WADI	MSL	PSM
RGAnomaly-onlyTime	0.8380	0.8558	0.8830	0.9508
RGAnomaly-onlyMetric	0.8600	0.8629	0.8890	0.9472
RGAnomaly-onlyAE	0.8657	0.8601	0.9180	0.9539
RGAnomaly-onlyVAE	0.8734	0.8578	0.9141	0.9561
RGAnomaly-noGAN	0.7966	0.6761	0.7309	0.8557
RGAnomaly	**0.8973**	**0.8635**	**0.9246**	**0.9775**

Table 4
Ablation studies of the overlapping windows.

Window	SWaT	WADI	MSL	PSM	Avg
overlapping=10	0.8724	0.8605	0.8987	0.9571	0.8972
overlapping=20	0.8834	**0.8696**	0.9146	0.9510	0.9047
overlapping=30	0.8442	0.8612	0.9204	0.9666	0.8981
overlapping=40	0.8858	0.7270	0.9100	0.9372	0.8650
non-overlapping	**0.8973**	0.8635	**0.9246**	**0.9775**	**0.9157**

The results of the ablation studies demonstrate that, in multivariate time series datasets, both temporal and metric features play a pivotal role in data reconstruction and anomaly detection. Currently, most anomaly detection models pay more attention to temporal feature extraction and ignore metric features. In fact, the implicit relationship between metrics can also be used to reflect the characteristics and distribution of data. In Fig. 3(a), the average anomaly detection effect of RGAnomaly-onlyMetric on the data set is even better than that of RGAnomaly-onlyTime. The result indicates that the combination of temporal and metric features can facilitate the comprehensive capture of complex patterns and anomalies in data, thereby enhancing the model's anomaly detection capability.

The experimental results of RGAnomaly-noGAN demonstrate that adversarial networks are highly effective in promoting anomaly detection. This is because GAN can significantly enhance the data reconstruction capability of the model, which is consistent with the findings of ADTransformer, USAD, and other models that employ an adversarial network structure. In addition, the discrimination loss plays a pivotal role in the calculation of the anomaly score for RGAnomaly. The Fig. 3(b) demonstrates a significant discrepancy between the complete RGAnomaly and the single data reconstruction model. This evidence substantiates the assertion that the generative adversarial network, when integrated with multiple data reconstruction models, can circumvent the constraints of the aforementioned single reconstruction model. This approach is more suitable for multivariate time series anomaly detection.

In this paper, we chose non-overlapping windows. To investigate the impact of overlapping windows on anomaly detection, we conducted ablation experiments with an overlap step size of 10, and the results are shown in the Table 4. Across the four datasets, the non-overlapping window approach achieved higher average F1 scores, with the exception of the WADI dataset, where it did not attain the optimal performance. We believe that, although overlapping windows generate more training data, the additional data does not necessarily improve

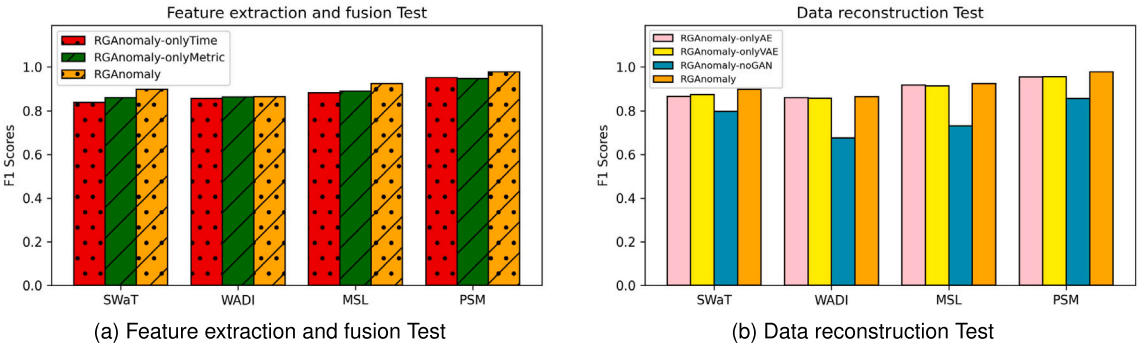


Fig. 3. The ablation tests for the validity of each part of the model, include feature extraction fusion experiment and data reconstruction experiment.

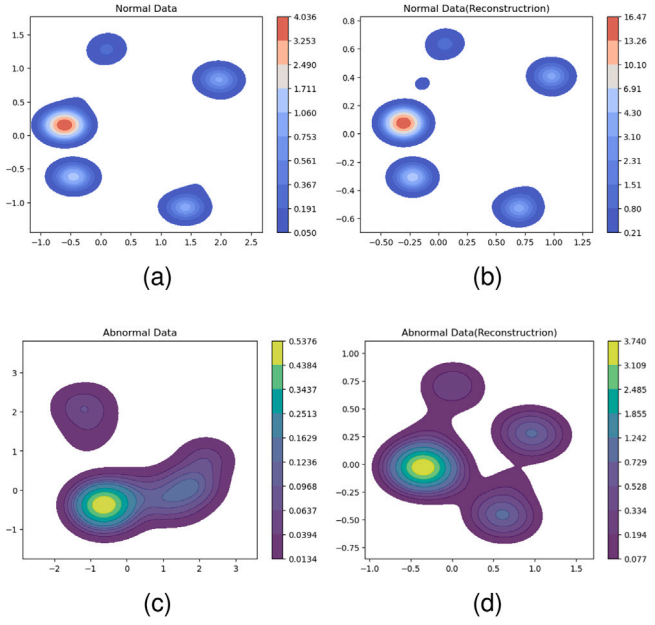


Fig. 4. Distribution of origin and reconstruction data in SWaT dataset. (a) and (c) show normal and abnormal data distributions, respectively, and (c) and (d) show corresponding reconstructed data distributions.

detection performance. This is because repeated data may lead the model to overfit certain high-frequency features, focusing excessively on repetitive local patterns. Furthermore, the repeated appearance of some anomalous information in the dataset could cause the model to shift towards an incorrect distribution. Finally, the repeated data introduced by overlapping windows increases computational overhead. Therefore, for the model in this study, using non-overlapping windows is a more effective choice.

5.4.3. Visual analysis

To better evaluate the model's performance, we visualized its data reconstruction capabilities and anomaly detection abilities. For data reconstruction, we aim for the model to accurately reconstruct normal data while exhibiting significant reconstruction differences for anomalous data. We visualized the data distribution by reconstructing both normal and anomalous data. The Fig. 4(a) and (b) display the reconstruction and original distributions of normal data, respectively. The two distributions are nearly identical, demonstrating that the model can effectively reconstruct normal data. In contrast, Fig. 4(c) and (d) show the visualization results for anomalous data, where the distributions differ markedly. Thus, the model's data reconstruction capability effectively distinguishes between normal and anomalous data.

Based on data reconstruction, RGAnomaly calculates the reconstruction error to obtain anomaly scores. Anomalous data exhibit higher reconstruction errors, leading to higher anomaly scores compared to normal data. By comparing the anomaly scores to a threshold δ, anomalies are detected. Fig. 5 presents the anomaly detection results on the PSM dataset. In both subfigures, the blue solid line represents the anomaly scores at different time points, the red dashed line indicates the threshold, and the green shaded regions highlight the practical anomalous areas. Fig. 5(a) demonstrates that the model accurately detects three discontinuous anomalies, where the anomaly scores at these time points significantly exceed the threshold, indicating a clear distinction between normal and anomalous data. In Fig. 5(b), the anomaly scores within the continuous anomaly region show a noticeable difference compared to those in the normal region, proving that the model can effectively identify continuous anomalies that deviate from normal data. These experimental results prove that the model is capable of effectively detecting various types of anomalies, including discontinuous point anomalies and continuous pattern anomalies.

5.4.4. Sensitivity analysis

Training dataset ratio sensitivity: In the natural IoT environment, obtaining sufficient and normal time series data for training is difficult due to unknown and complex dynamic anomalies. Therefore, the anomaly detection model still needs to have anomaly detection ability in the case of insufficient data training. We split the four datasets into 25%, 50%, 75% and 100%, the experimental results are shown in Fig. 6. The RGAnomaly still exhibits a certain level of anomaly detection capability on the MSL and PSM datasets with only 25% of the training data. However, its performance on the SWaT and WADI datasets is relatively poorer, particularly in the case of the SWaT dataset. This is related to the data distribution and anomaly types of the datasets. The SWaT and WADI datasets are analogous water treatment distribution datasets, with more training data than the others. As the ratio of training data increases, particularly when it reaches 50%, the model demonstrates good anomaly detection capability, proving that it can effectively learn the normal data distribution and perform time series data reconstruction. Although there is a slight decline in performance for some datasets with further increases in the training data ratio, this is due to the presence of unknown anomalies within the datasets. Nevertheless, the RGAnomaly demonstrates a clear upward trend across all four datasets.

Noise sensitivity: Due to the complex IoT environments, real-world data often contains noise. To better simulate practical application scenarios, we added white Gaussian noise and Gaussian noise to the dataset. For white Gaussian noise, the noise decibel decreases from

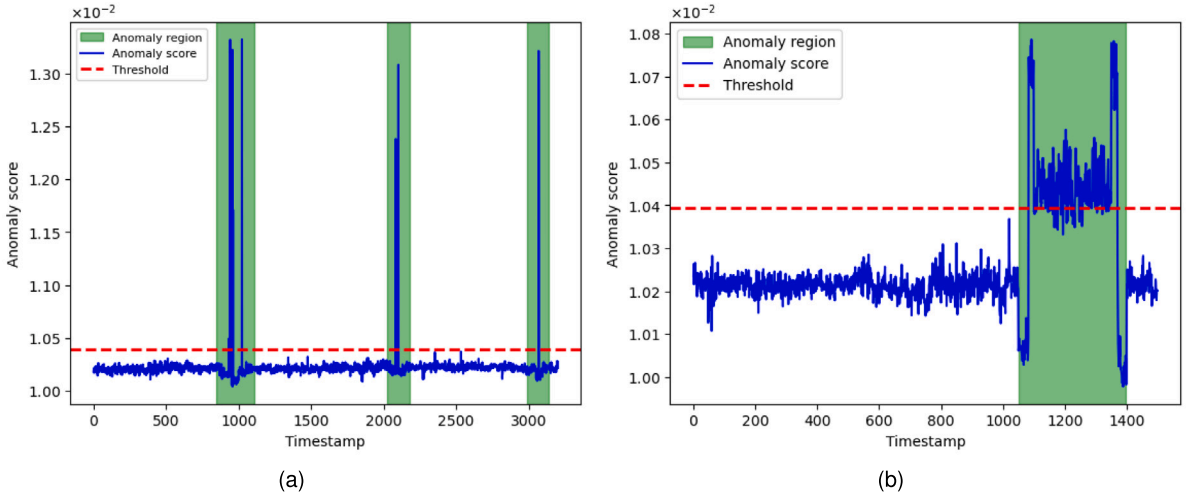


Fig. 5. Anomaly detection results on the PSM dataset. The blue solid line represents the anomaly score, the green shadow represents the anomalies, and the red dashed line represents the threshold.

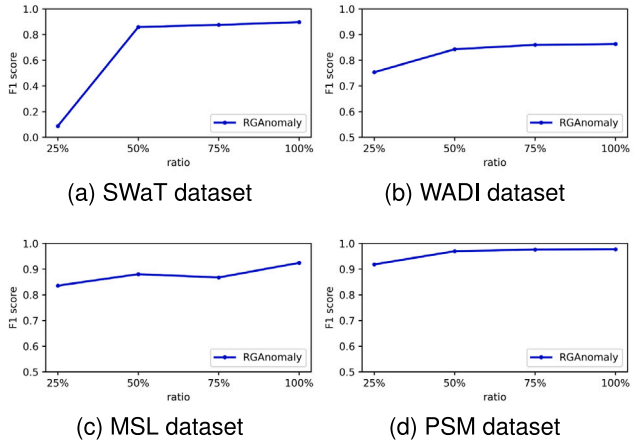


Fig. 6. Sensitivity test of the dataset ratio.

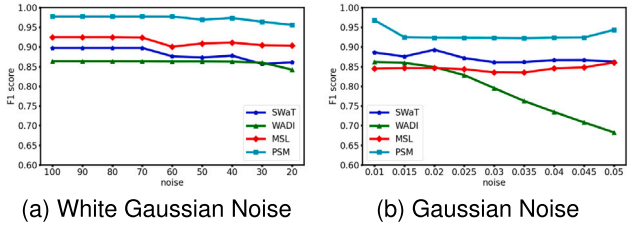


Fig. 7. Sensitivity test of the dataset noise.

100 dB, and the lower the decibel value, the greater the impact on the original data. As shown in Fig. 7(a), from 100 dB to 60 dB, the F1 scores for the four datasets remain consistently stable. As the decibel level decreases, the noise increasingly affects the original data, resulting in a decline in anomaly detection performance. However, the decline is not substantial. As for the Gaussian noise, the standard deviation of the noise increases from 0.01 and the effect on the data increases gradually. The test results are shown in Fig. 7(b), where it can be seen that the model performs more stably on most datasets, and only on the WADI dataset there is a more pronounced performance decline. Therefore, RGAnomaly demonstrates robustness when dealing with noisy environments. We believe that, compared to random noise, the feature extraction and fusion module focuses more on the key information within the data, such as feature correlations and latent patterns, thereby mitigating the interference caused by noise. Furthermore, the unique generative adversarial network employed in our model effectively learns the primary distribution characteristics of the data, which may help reduce the influence of the noise distribution while preserving the fundamental properties of the data.

Model parameters sensitivity: In the RGAnomaly model, the most significant model parameters include the window size, the number of EncoderBlock, the dimension of the attention vector(*d_model*), and the hidden layer dimension of AE, VAE, and GAN (for the sake of convenience, the three are considered a single variable). In order to analyze the effect of different parameter on model performance, a comparative experiment was conducted on these four parameters, with the results in Fig. 8. Fig. 8(a) demonstrates the impact of sliding window size on anomaly detection performance. Overall, the model exhibits good detection capability across different window size parameters. Moreover, it is not the case that the larger or smaller the window size, the better the corresponding performance. Firstly, smaller windows are more susceptible to noise because they provide less information, making distinguishing between noise and genuine anomaly signals difficult. Secondly, more oversized windows might contain too much information, making it challenging for the model to identify short-term anomaly patterns accurately. Fig. 8(b) demonstrates that the different encoder layer parameters show relatively stable detection performance on most of the datasets. It is notable that increasing the number of layers does not necessarily enhance the model's performance, especially on the SWaT dataset. Subsequent feature extraction and fusion can disrupt the continuity of temporal features after one round. This is also why Interfusion chooses to extract temporal features first and then extract metric features on top of them. Fig. 8(c) and Fig. 8(d) investigate the influence of model dimensions on anomaly detection performance. These figures demonstrate that RGAnomaly is not particularly sensitive to *d_model* and hidden dimensions. The result reflects the robustness of the model, indicating that RGAnomaly can provide excellent anomaly detection capabilities even without the most optimal parameter settings.

6. Conclusion

This paper proposes RGAnomaly, a novel data reconstruction-based generative adversarial network. By constructing a feature extraction

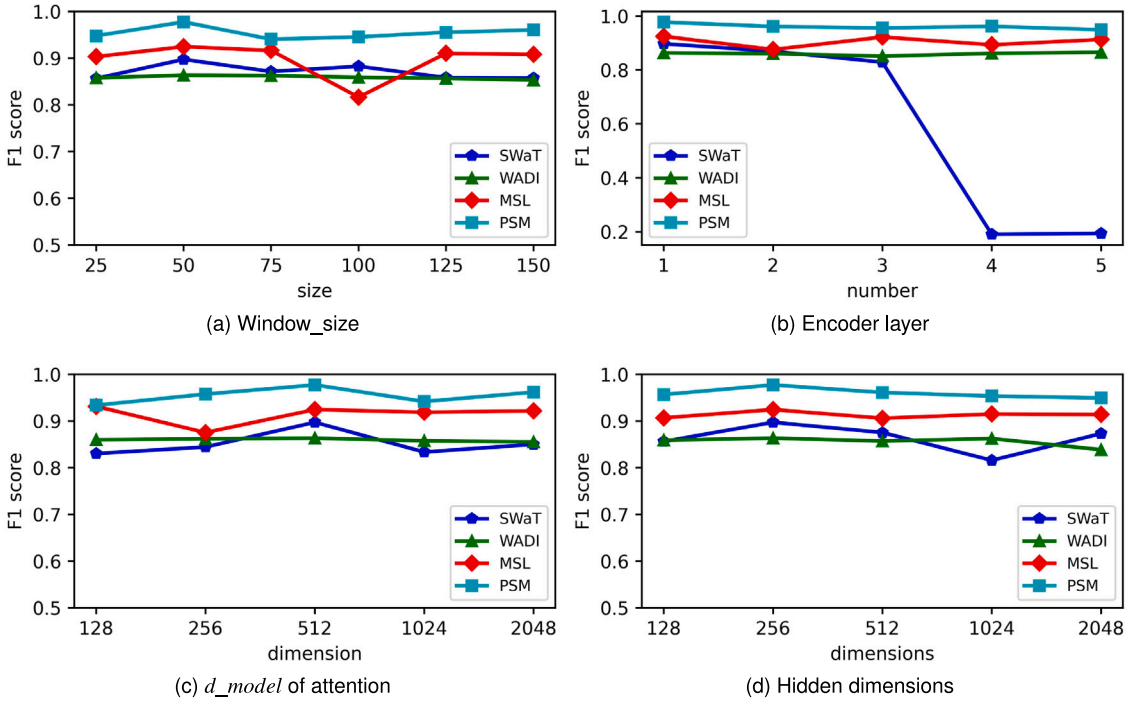


Fig. 8. Sensitivity test of the RGAnomaly's parameters.

and fusion module, our model addresses the insufficiencies in feature extraction and fusion present in existing methods. Integrating multiple data reconstruction models within the generative adversarial network enhances the model's anomaly detection capability, mitigating the instability issues associated with single data reconstruction models. The comprehensive experimental results not only demonstrate the model's advanced anomaly detection capabilities but also validate the effectiveness of each module and the model's robustness to factors such as noise. Consequently, we believe that RGAnomaly is suitable for multivariate time series anomaly detection in IoT environments.

Furthermore, since RGAnomaly detects anomalies by learning the features and data distributions of multivariate time series, it can theoretically be trained and applied to various IoT environments as long as the data are multivariate time series with recognizable patterns. Although this is a theoretical assumption, the comparative experiments in this paper utilized datasets from three different IoT environments, which serve as examples to demonstrate the model's capability to adapt to different IoT applications. However, we also acknowledge that the deployment of RGAnomaly, as a deep learning model, may face limitations in terms of computational resources and energy consumption. Therefore, the model is more suitable for deployment in control centers or edge computing servers with anomaly detection requirements. We believe that RGAnomaly may have certain limitations when dealing with anomalies exhibiting long-term dependencies. In addition, it may face challenges in scenarios with insufficient training data or sparse anomalies. For high-dimensional time series data in IoT environments, the model might be affected by the curse of dimensionality, and its scalability has room for further improvement.

These challenges mentioned above provide valuable directions for future research. In future work, we plan to explore anomaly detection methods that combine dimensionality reduction with contrastive learning. Dimensionality reduction has the potential to simplify high-dimensional data, reduce computational complexity, while still retaining key information. This can be used to perform contrastive learning with the original data, allowing the model to focus more on the critical features or data distributions within the time series. In addition, we plan to explore modeling approaches for long-term dependencies and online learning frameworks to further enhance the proposed method.

CRediT authorship contribution statement

Cheng Qian: Writing – original draft, Visualization, Software, Investigation, Data curation. **Wenzhong Tang:** Supervision, Project administration, Methodology. **Yanyang Wang:** Writing – review & editing, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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